

Constructing Derivatives: The Pattern-Book Functions

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Outline (75 minutes)

1. **The Pattern-Book Functions** – Review and motivation (10 min)
2. **Derivatives via Limit Definition** – Rigorous derivations (25 min)
3. **The Complete Derivative Table** – All pattern-book derivatives (15 min)
4. **Higher-Order Derivatives** – Second, third, and beyond (15 min)
5. **Finite-Difference Approximations** – Numerical methods (10 min)

1 The Pattern-Book Functions

1.1 What Are Pattern-Book Functions?

Definition: The **pattern-book functions** are the fundamental building blocks from which more complex functions are constructed. They are the “characters” that appear repeatedly in mathematical modeling (see Lecture 1).

The nine pattern-book functions:

1. **Exponential:** e^x
2. **Logarithm:** $\ln(x)$
3. **Sinusoid:** $\sin(x)$ (and $\cos(x)$)
4. **Square:** x^2
5. **Identity:** x
6. **One:** 1 (constant)
7. **Reciprocal:** $1/x$ or x^{-1}
8. **Gaussian:** e^{-x^2} (bell curve)
9. **Sigmoid:** $\frac{1}{1 + e^{-x}}$ (S-curve)

1.2 Why These Functions?

Universality: These functions appear across all sciences:

- Exponential and logarithm: growth, decay, radioactivity, finance, information theory
- Sinusoid: oscillations, waves, periodic phenomena
- Square, identity, one, reciprocal: scaling laws, geometry, linear and power relationships
- Gaussian: probability distributions, error analysis
- Sigmoid: population limits, neural networks, dose-response

Our goal: Find the derivative of each pattern-book function so we can analyze rates of change in any model built from them.

2 Derivatives via Limit Definition

2.1 The Power Rule: Proof

We first state a fact we will use.

Lemma 2.1 (Binomial expansion). *For any positive integer n ,*

$$(x + h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \binom{n}{3}x^{n-3}h^3 + \cdots + h^n.$$

(The binomial theorem gives this expansion; $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.)

Theorem 2.2 (Power Rule). *For any positive integer n :*

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

Proof (step-by-step):

Step 1. By definition, the derivative is the limit of the difference quotient:

$$\frac{d}{dx}(x^n) = \lim_{h \rightarrow 0} \frac{(x + h)^n - x^n}{h}.$$

Step 2. By Lemma 2.1 (binomial expansion), we replace $(x + h)^n$:

$$(x + h)^n - x^n = \left(x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n \right) - x^n = nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n.$$

Step 3. Divide by h (for $h \neq 0$):

$$\frac{(x + h)^n - x^n}{h} = \frac{nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots + h^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots + h^{n-1}.$$

Step 4. Take the limit as $h \rightarrow 0$. Every term after the first contains a factor h , so those terms go to zero. Thus

$$\lim_{h \rightarrow 0} \left[nx^{n-1} + \binom{n}{2} x^{n-2} h + \cdots + h^{n-1} \right] = nx^{n-1} + 0 + \cdots + 0 = nx^{n-1}.$$

Step 5. Therefore $\frac{d}{dx}(x^n) = nx^{n-1}$. $\square$

2.2 The Exponential Function: Proof

We need one fact about the number e .

Lemma 2.3 (Limit for the exponential).

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1.$$

(One can show this using the series $e^h = 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \cdots$; then $\frac{e^h - 1}{h} = 1 + \frac{h}{2!} + \frac{h^2}{3!} + \cdots$, and as $h \rightarrow 0$ this tends to 1.)

Theorem 2.4.

$$\frac{d}{dx}(e^x) = e^x.$$

The exponential function is its own derivative!

Proof (step-by-step):

Step 1. By definition:

$$\frac{d}{dx}(e^x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h}.$$

Step 2. Use the property $e^{x+h} = e^x \cdot e^h$ (exponent rule). So

$$\frac{e^{x+h} - e^x}{h} = \frac{e^x \cdot e^h - e^x}{h} = e^x \cdot \frac{e^h - 1}{h}.$$

Step 3. As $h \rightarrow 0$, e^x does not depend on h , so it stays constant. By Lemma 2.3, $\frac{e^h - 1}{h} \rightarrow 1$. So

$$\lim_{h \rightarrow 0} \left(e^x \cdot \frac{e^h - 1}{h} \right) = e^x \cdot 1 = e^x.$$

Step 4. Therefore $\frac{d}{dx}(e^x) = e^x$. $\square$

Remark: This property—being equal to its own derivative—is what defines the number $e \approx 2.71828$. No other base has this property.

2.3 General Exponential Functions

We will use the chain rule (derivative of a composition). We state it here as a lemma; it will be proved in a later lecture.

Lemma 2.5 (Chain rule). *If $f(x) = g(u(x))$, then $\frac{df}{dx} = \frac{dg}{du} \cdot \frac{du}{dx}$ (at any x where the right-hand side makes sense).*

Proposition 2.6. *For any positive constant a :*

$$\frac{d}{dx}(a^x) = a^x \ln(a).$$

Proof (step-by-step):

Step 1. Write a^x in terms of e . By definition of the natural logarithm, $a = e^{\ln a}$, so $a^x = (e^{\ln a})^x = e^{x \ln a}$.

Step 2. So $a^x = g(u(x))$ where $g(u) = e^u$ and $u(x) = x \ln a$. By Lemma 2.5 (chain rule), $\frac{d}{dx}(a^x) = \frac{d}{du}(e^u) \cdot \frac{du}{dx}$.

Step 3. We know $\frac{d}{du}(e^u) = e^u$ and $\frac{du}{dx} = \ln a$. So $\frac{d}{dx}(a^x) = e^{x \ln a} \cdot \ln a = a^x \ln a$.

Step 4. Therefore $\frac{d}{dx}(a^x) = a^x \ln a$. $\square$

Special cases:

- $\frac{d}{dx}(2^x) = 2^x \ln 2 \approx 0.693 \cdot 2^x$
- $\frac{d}{dx}(10^x) = 10^x \ln 10 \approx 2.303 \cdot 10^x$

2.4 The Natural Logarithm: Proof

We need one limit fact.

Lemma 2.7 (Limit for the logarithm).

$$\lim_{u \rightarrow 0} \frac{\ln(1+u)}{u} = 1.$$

(This can be justified using the series $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \dots$; then $\frac{\ln(1+u)}{u} = 1 - \frac{u}{2} + \frac{u^2}{3} - \dots \rightarrow 1$ as $u \rightarrow 0$.)

Theorem 2.8.

$$\frac{d}{dx}(\ln x) = \frac{1}{x} \quad \text{for } x > 0.$$

Proof (step-by-step):

Step 1. By definition:

$$\frac{d}{dx}(\ln x) = \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln(x)}{h}.$$

Step 2. Use the logarithm rule $\ln A - \ln B = \ln(A/B)$. So

$$\ln(x+h) - \ln(x) = \ln\left(\frac{x+h}{x}\right) = \ln\left(1 + \frac{h}{x}\right).$$

Thus the difference quotient is

$$\frac{1}{h} \ln\left(1 + \frac{h}{x}\right).$$

Step 3. Let $u = h/x$. Then $h = ux$, and when $h \rightarrow 0$, we have $u \rightarrow 0$.

So

$$\frac{1}{h} \ln\left(1 + \frac{h}{x}\right) = \frac{1}{ux} \ln(1+u) = \frac{1}{x} \cdot \frac{\ln(1+u)}{u}.$$

Step 4. As $h \rightarrow 0$, $u \rightarrow 0$. The factor $\frac{1}{x}$ is constant. By Lemma 2.7, $\frac{\ln(1+u)}{u} \rightarrow 1$. So

$$\lim_{h \rightarrow 0} \frac{1}{x} \cdot \frac{\ln(1+u)}{u} = \frac{1}{x} \cdot 1 = \frac{1}{x}.$$

Step 5. Therefore $\frac{d}{dx}(\ln x) = \frac{1}{x}$. $\square$

2.5 Sine and Cosine: Proofs

We need two facts: the angle addition formulas and two limits.

Lemma 2.9 (Angle addition formulas). *For any x and h :*

$$\sin(x+h) = \sin x \cos h + \cos x \sin h, \quad \cos(x+h) = \cos x \cos h - \sin x \sin h.$$

Lemma 2.10 (Fundamental trig limits).

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0.$$

(These hold when angles are in radians.)

Theorem 2.11.

$$\frac{d}{dx}(\sin x) = \cos x \quad \text{and} \quad \frac{d}{dx}(\cos x) = -\sin x.$$

Proof for sine (step-by-step):

Step 1. By definition:

$$\frac{d}{dx}(\sin x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}.$$

Step 2. By Lemma 2.9 (angle addition), $\sin(x + h) = \sin x \cos h + \cos x \sin h$. So

$$\sin(x + h) - \sin x = \sin x \cos h + \cos x \sin h - \sin x = \sin x(\cos h - 1) + \cos x \sin h.$$

Step 3. Divide by h :

$$\frac{\sin(x + h) - \sin x}{h} = \sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h}.$$

Step 4. Take the limit as $h \rightarrow 0$. The terms $\sin x$ and $\cos x$ do not depend on h . By Lemma 2.10, $\frac{\cos h - 1}{h} \rightarrow 0$ and $\frac{\sin h}{h} \rightarrow 1$. So

$$\lim_{h \rightarrow 0} \left(\sin x \cdot \frac{\cos h - 1}{h} + \cos x \cdot \frac{\sin h}{h} \right) = \sin x \cdot 0 + \cos x \cdot 1 = \cos x.$$

Step 5. Therefore $\frac{d}{dx}(\sin x) = \cos x$. $\square$

Proof for cosine (step-by-step):

Step 1. By definition: $\frac{d}{dx}(\cos x) = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$.

Step 2. By Lemma 2.9, $\cos(x + h) = \cos x \cos h - \sin x \sin h$. So $\cos(x + h) - \cos x = \cos x(\cos h - 1) - \sin x \sin h$.

Step 3. Divide by h : $\frac{\cos(x+h) - \cos x}{h} = \cos x \cdot \frac{\cos h - 1}{h} - \sin x \cdot \frac{\sin h}{h}$.

Step 4. Take the limit as $h \rightarrow 0$: $\cos x \cdot 0 - \sin x \cdot 1 = -\sin x$.

Step 5. Therefore $\frac{d}{dx}(\cos x) = -\sin x$. $\square$

Important note: These formulas require x to be measured in **radians**. If x is in degrees, conversion factors appear.

3 The Complete Derivative Table

3.1 Pattern-Book Function Derivatives

Function	Derivative	Domain
c (constant)	0	all x
x^n	nx^{n-1}	depends on n
e^x	e^x	all x
a^x	$a^x \ln a$	all x
$\ln x$	$\frac{1}{x}$	$x > 0$
$\log_a x$	$\frac{1}{x \ln a}$	$x > 0$
$\sin x$	$\cos x$	all x
$\cos x$	$-\sin x$	all x
$\tan x$	$\sec^2 x$	$x \neq \frac{\pi}{2} + n\pi$

3.2 Extended Power Rule Examples

The power rule $\frac{d}{dx}(x^n) = nx^{n-1}$ works for *any* real exponent:

Fractional exponents:

$$\begin{aligned}\frac{d}{dx}(\sqrt{x}) &= \frac{d}{dx}(x^{1/2}) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}} \\ \frac{d}{dx}(\sqrt[3]{x}) &= \frac{d}{dx}(x^{1/3}) = \frac{1}{3}x^{-2/3} = \frac{1}{3\sqrt[3]{x^2}}\end{aligned}$$

Negative exponents:

$$\begin{aligned}\frac{d}{dx}\left(\frac{1}{x}\right) &= \frac{d}{dx}(x^{-1}) = -x^{-2} = -\frac{1}{x^2} \\ \frac{d}{dx}\left(\frac{1}{x^3}\right) &= \frac{d}{dx}(x^{-3}) = -3x^{-4} = -\frac{3}{x^4}\end{aligned}$$

Irrational exponents:

$$\frac{d}{dx}(x^\pi) = \pi x^{\pi-1}$$

3.3 The Derivative of the Gaussian

Function: $g(x) = e^{-x^2}$ (the bell curve)

Using the chain rule: Let $u = -x^2$, so $g(x) = e^u$.

$$\frac{dg}{dx} = \frac{d}{du}(e^u) \cdot \frac{du}{dx} = e^u \cdot (-2x) = -2xe^{-x^2}$$

Key property: $g'(0) = 0$, confirming the peak at $x = 0$.

For $x > 0$: $g'(x) < 0$ (decreasing)

For $x < 0$: $g'(x) > 0$ (increasing)

4 Higher-Order Derivatives

4.1 Definition and Notation

Definition 4.1. The **second derivative** of f is the derivative of the first derivative:

$$f''(x) = \frac{d}{dx} \left(\frac{df}{dx} \right) = \frac{d^2 f}{dx^2}$$

Higher derivatives:

$$f'''(x) = \frac{d^3 f}{dx^3} \quad (\text{third derivative})$$

$$f^{(4)}(x) = \frac{d^4 f}{dx^4} \quad (\text{fourth derivative})$$

$$f^{(n)}(x) = \frac{d^n f}{dx^n} \quad (n\text{th derivative})$$

4.2 Example: Power Functions

Let $f(x) = x^5$:

$$f'(x) = 5x^4$$

$$f''(x) = 20x^3$$

$$f'''(x) = 60x^2$$

$$f^{(4)}(x) = 120x$$

$$f^{(5)}(x) = 120$$

$$f^{(6)}(x) = 0$$

Pattern: For $f(x) = x^n$, the n th derivative is $n!$ (a constant), and all higher derivatives are zero.

4.3 Example: Exponential Function

Let $f(x) = e^x$:

$$f'(x) = e^x$$

$$f''(x) = e^x$$

$$f'''(x) = e^x$$

$$\vdots$$

$$f^{(n)}(x) = e^x$$

Remarkable property: The exponential function equals all of its derivatives!

4.4 Example: Sine Function

Let $f(x) = \sin x$:

$$\begin{aligned}f'(x) &= \cos x \\f''(x) &= -\sin x \\f'''(x) &= -\cos x \\f^{(4)}(x) &= \sin x\end{aligned}$$

Pattern: The derivatives of sine cycle with period 4:

$$\sin x \rightarrow \cos x \rightarrow -\sin x \rightarrow -\cos x \rightarrow \sin x \rightarrow \cdots$$

4.5 Physical Interpretation: Kinematics

For position $x(t)$:

$$\begin{aligned}x(t) &= \text{position} \\x'(t) &= v(t) = \text{velocity} \\x''(t) &= a(t) = \text{acceleration} \\x'''(t) &= j(t) = \text{jerk (rate of change of acceleration)}\end{aligned}$$

Example: Free fall with $x(t) = x_0 + v_0t - \frac{1}{2}gt^2$

$$\begin{aligned}x'(t) &= v_0 - gt \\x''(t) &= -g \quad (\text{constant acceleration}) \\x'''(t) &= 0 \quad (\text{no jerk})\end{aligned}$$

5 Finite-Difference Approximations

5.1 When Formulas Aren't Available

The problem: Sometimes we have a function defined by data or a complex formula that's hard to differentiate symbolically.

The solution: Approximate the derivative numerically.

5.2 Forward Difference

Approximation:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

This is just the difference quotient with small (but nonzero) h .

Error: $O(h)$ — halving h roughly halves the error.

5.3 Central Difference (Better)

Approximation:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

Why better? This is symmetric around x , which cancels first-order errors.

Error: $O(h^2)$ — halving h reduces error by factor of 4.

5.4 Example: Approximating $\frac{d}{dx}(e^x)$ at $x = 1$

Exact value: $e^1 \approx 2.71828$

Forward difference:

h	Approximation	Error
0.1	$(e^{1.1} - e^1)/0.1 = 2.8588$	0.1405
0.01	$(e^{1.01} - e^1)/0.01 = 2.7319$	0.0136
0.001	2.7196	0.0014

Central difference:

h	Approximation	Error
0.1	$(e^{1.1} - e^{0.9})/0.2 = 2.7228$	0.0045
0.01	2.71833	0.00005
0.001	2.718282	0.0000005

Central difference converges much faster!

5.5 Numerical Pitfall: Too Small h

Warning: Making h extremely small doesn't always help!

Reason: Computers store numbers with limited precision (~ 15 decimal digits). When h is tiny, $f(x+h) - f(x)$ involves subtracting nearly equal numbers, causing **catastrophic cancellation**.

Practical advice: Use $h \approx 10^{-5}$ to 10^{-8} , and verify by checking that results are stable as h varies.

6 Summary

6.1 Key Derivative Formulas

$$\begin{aligned}\frac{d}{dx}(x^n) &= nx^{n-1} \\ \frac{d}{dx}(e^x) &= e^x \\ \frac{d}{dx}(\ln x) &= \frac{1}{x} \\ \frac{d}{dx}(\sin x) &= \cos x \\ \frac{d}{dx}(\cos x) &= -\sin x\end{aligned}$$

6.2 Practice Exercises

Exercise 1: Use the limit definition to prove that $\frac{d}{dx}(x^4) = 4x^3$.

Exercise 2: Verify numerically that $\frac{d}{dx}(\ln x)$ at $x = 2$ equals 0.5 using central differences with $h = 0.01$.

Exercise 3: Find all derivatives of $f(x) = x^6$ up to $f^{(7)}(x)$.

Exercise 4: For $f(x) = e^{-x}$, find $f'(x)$, $f''(x)$, and $f'''(x)$. What pattern emerges?

Exercise 5: A car's position is $x(t) = 10t - t^2 + 0.1t^3$. Find velocity, acceleration, and jerk as functions of time.

Worked Solutions

Exercise 1 (Solution): Prove $\frac{d}{dx}(x^4) = 4x^3$ using the limit definition.

Step 1. Difference quotient: $\frac{(x+h)^4 - x^4}{h}$.

Step 2. Expand $(x+h)^4 = x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4$, so $(x+h)^4 - x^4 = 4x^3h + 6x^2h^2 + 4xh^3 + h^4$.

Step 3. Divide by h : $\frac{(x+h)^4 - x^4}{h} = 4x^3 + 6x^2h + 4xh^2 + h^3$.

Step 4. As $h \rightarrow 0$, the limit is $4x^3$. So $\frac{d}{dx}(x^4) = 4x^3$. $\square$

Exercise 2 (Solution): Central difference approximation: $f'(2) \approx \frac{\ln(2.01) - \ln(1.99)}{0.02}$. We have $\ln(2.01) \approx 0.6981$, $\ln(1.99) \approx 0.6881$, so $\frac{0.6981 - 0.6881}{0.02} = \frac{0.01}{0.02} = 0.5$. The exact derivative is $\frac{1}{2} = 0.5$. $\checkmark$ $\square$

Exercise 3 (Solution): For $f(x) = x^6$: $f'(x) = 6x^5$, $f''(x) = 30x^4$, $f'''(x) = 120x^3$, $f^{(4)}(x) = 360x^2$, $f^{(5)}(x) = 720x$, $f^{(6)}(x) = 720$, $f^{(7)}(x) = 0$. $\square$

Exercise 4 (Solution): For $f(x) = e^{-x}$: $f'(x) = -e^{-x}$, $f''(x) = e^{-x}$, $f'''(x) = -e^{-x}$. The pattern: $f^{(n)}(x) = (-1)^n e^{-x}$. Each derivative alternates sign; the magnitude is always e^{-x} . $\square$

Exercise 5 (Solution): $x(t) = 10t - t^2 + 0.1t^3$. Velocity: $v(t) = x'(t) = 10 - 2t + 0.3t^2$. Acceleration: $a(t) = v'(t) = -2 + 0.6t$. Jerk: $j(t) = a'(t) = 0.6$. $\square$